



PRODUCT NAME

Premium Arm Pain Formula

Description

A cost-effective cream designed to provide anti-inflammatory and pain relief benefits with fast absorption. Ideal for soothing arm pain and minimizing discomfort.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	24 months
Batch Size:	50 kg to 100 kg
Processing Time:	2 hours
Temperature:	Processing temperature 75°C
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, heating source, high-speed mixer
Certification:	Manufacture in GMP-certified facility

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Glycerin	Glycerin	10.0%	Humectant
3	Cetyl Alcohol	Cetyl Alcohol	5.0%	Emollient, Thickener
4	Glyceryl Stearate	Glyceryl Stearate	6.0%	Emulsifying agent
5	Menthol	Menthol	2.0%	Cooling agent, Pain relief
6	Methyl Salicylate	Methyl Salicylate	4.0%	Anti-inflammatory, Pain relief
7	Eucalyptus Oil	Eucalyptus Globulus Leaf Oil	4.0%	Pain relief, Anti-inflammatory
8	Stearic Acid	Stearic Acid	5.0%	Thickener
9	Triethanolamine	Triethanolamine	2.0%	pH adjuster
10	Phenoxyethanol	Phenoxyethanol	2.0%	Preservative
Total			100%	

Manufacturing Process

Phase 1: Heat Phase

1. Combine Water, Glycerin, and Triethanolamine in a large mixing vessel.
2. Heat to 75°C with continuous stirring until fully dissolved.

Phase 2: Oil Phase

1. In a separate vessel, mix Cetyl Alcohol, Glyceryl Stearate, Stearic Acid, Menthol, and Methyl Salicylate.
2. Heat to 75°C until homogenized.

Phase 3: Emulsification Phase

1. Add the Oil Phase into the Water Phase slowly with high-speed stirring.
2. Continue mixing for 10 minutes until a uniform emulsion is formed.

Phase 4: Cooling Phase

1. Allow the mixture to cool down to 40°C while reducing stirring speed.
2. Add Eucalyptus Oil and Phenoxyethanol, mix until fully incorporated.

Required Equipment

Mixing vessel, heating source, high-speed mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Manufacture in GMP-certified facility

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50 kg to 100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of cream onto the affected arm area. Gently massage until absorbed. Use up to three times daily for optimal relief.